

Proper Forcing Axiom

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1 Basics

1.1 Introducing Properness

- I. We will assume that every poset $\mathbb{P}$ is defined on $|\mathbb{P}|$.
- II. **Definition.** Let $\theta > 2^{|\mathbb{P}|}$ be regular and let $M \prec (H_\theta, \in, \mathbb{P})$ be countable. We say that $p \in \mathbb{P}$ is $M^\mathbb{P}$ -generic if

$$p \Vdash (\forall D \in M)(D \text{ pre-dense in } \mathbb{P} \implies G \cap D \cap M \neq \emptyset).$$

- III. **Remark.** In other words, if $k : \overline{M} \cong M \prec H_\theta$ is the anti-collapse, with $\overline{\mathbb{P}} := k^{-1}(\mathbb{P})$, we require

$$p \Vdash k^{-1}[G] \rightsquigarrow \overline{M}^{\overline{\mathbb{P}}}.$$

- IV. **Remark.** For $M \prec (H_\theta, \in, \mathbb{P})$ and $G \rightsquigarrow V^\mathbb{P}$, we use the following notation:

$$M[G] := \{\dot{x}^G : \dot{x} \in M^\mathbb{P}\},$$

where $M^\mathbb{P} = (V^\mathbb{P})^M = V^\mathbb{P} \cap M$.

- V. **Proposition.** Let $\theta > 2^{|\mathbb{P}|}$ be regular, let $M \prec (H_\theta, \in, \mathbb{P})$ be countable, and let $p \in \mathbb{P}$. Then the following are equivalent.
- a. p is $M^\mathbb{P}$ -generic.
 - b. For every $D \in M$ pre-dense in $\mathbb{P}$, we have $(\forall q \leq p)(\exists r \in D \cap M)(q \parallel r)$.
 - c. $p \Vdash M[G] \cap V = M$
 - d. $p \Vdash M[G] \cap \text{Ord} = M \cap \text{Ord}$

Proof.

Equivalence (a $\Leftrightarrow$ b)

1° Let $D \in M$ be pre-dense in $\mathbb{P}$. We need to verify

$$p \Vdash G \cap D \cap M \iff (\forall q \leq p)(\exists r \in D \cap M)(q \parallel r).$$

2° This is obvious since the right hand side is equivalent to stating that

$$(D \cap M) \cup \{r \in \mathbb{P} : r \perp p\}$$

is pre-dense in $\mathbb{P}$.

Implication (a $\Rightarrow$ c)

3° Let $G \rightsquigarrow V^\mathbb{P}$ such that $G \ni p$.

4° $M \subseteq M[G]$ is clear.

5° Let $\dot{x} \in M^\mathbb{P}$ such that $\dot{x}^G \in V$. We want to show $\dot{x}^G \in M$.

6° Let $\dot{y}$ be a name such that:

$$q \Vdash \dot{x} \in V \implies q \Vdash \dot{y} = \dot{x},$$

$$q \Vdash \dot{x} \notin V \implies q \Vdash \dot{y} = 0.$$

$\dot{y}$ can be chosen in M .

7° Since $\dot{x}^G \in V$, there is $q \in G$ such that $q \Vdash \dot{x} \in V$. This implies $q \Vdash \dot{x} = \dot{y}$ and consequently, $\dot{y}^G = \dot{x}^G$.

8° $D := \{r : r \parallel \dot{y}\} \in M$ is dense, so there is $r \in D \cap M \cap G$.

9° $\dot{y}^G$ is then definable as that value of $\dot{y}$ decided by r , so $\dot{y}^G \in M$.

Implication (c $\Rightarrow$ d)

10° Obvious.

Implication (d $\Rightarrow$ a)

11° Let $A \in M$ be an arbitrary maximal anti-chain.

12° Let $(q_\alpha : \alpha < \gamma)$ be an enumeration of A in M .

13° Let $\dot{x} \in M^\mathbb{P}$ be such that $q_\alpha \Vdash \dot{x} = \alpha$ for every $\alpha < \gamma$.

14° Suppose that there is $G \ni p$ such that $G \cap A \cap M = \emptyset$.

15° Then $\alpha := \dot{x}^G \in M[G]$, while $q_\alpha \in G \cap A$.

16° On the other hand, we have $q_\alpha \notin M$, which has for a consequence that $\alpha \notin M$.

17° This contradicts d.

□

VI. Proposition. Let $\theta > 2^{|\mathbb{P}|}$ and let $M \prec (H_\theta, \in, \mathbb{P})$ be countable. Then $\Vdash M[G] \prec H_\theta^V[G] = H_\theta^{V[G]}$.

Proof.

1° Let $G \rightsquigarrow V^\mathbb{P}$. We want to show $M[G] \prec H_\theta^{V[G]}$. We apply Tarski-Vaught test.

2° Suppose that $H_\theta^{V[G]} \models \exists y \phi(\dot{x}^G, y)$, where $\dot{x} \in M^\mathbb{P}$. We want to find $\dot{y} \in M^\mathbb{P}$ such that $H_\theta^{V[G]} \models \phi(\dot{x}^G, \dot{y}^G)$.

3° We have

$$H_\theta^V \models (\Vdash \exists y (\exists z \phi(\dot{x}, z) \implies \phi(\dot{x}, y))).$$

4° By maximality,

$$H_\theta^V \models \exists \dot{y} (\Vdash \exists z \phi(\dot{x}, z) \implies \phi(\dot{x}, \dot{y})).$$

5° By elementarity, there is $\dot{y} \in M$ such that

$$H_\theta^V \models (\Vdash \exists z \phi(\dot{x}, z) \implies \phi(\dot{x}, \dot{y})).$$

6° Thus, $H_\theta^{V[G]} = H_\theta^V[G] \models \phi(\dot{x}^G, \dot{y}^G)$, with $\dot{y}^G \in M[G]$.

□

VII. **Remark.** For $G \rightsquigarrow V^\mathbb{P}$, model $M[G] \prec H_\theta^{V[G]}$ computes $M[G] \cap V \prec H_\theta^V$ as its ground. In general, we only have $M \subseteq M[G] \cap V$. The equality holds if and only if G contains an $M^\mathbb{P}$ -generic condition. For example, if $\mathbb{P} := \text{Col}(\omega, \omega_1^V)$, then $\omega_1^V \subseteq M[G]$, so $M[G] \cap V \supseteq \omega_1^V$ cannot be contained in the countable M .

VIII. **Proposition.** Let $\mathbb{P}$ be a poset and let $p \in \mathbb{P}$. For every regular $\theta > 2^{2^{|\mathbb{P}|}}$, the following are equivalent.

- a. $\{M \prec (H((2^{|\mathbb{P}|})^+), \in, \mathbb{P}, p) : |M| = \omega \wedge (\exists q \leq p)(q \text{ is } M^\mathbb{P}\text{-generic})\}$ is club in $[H((2^{|\mathbb{P}|})^+)]^\omega$.
- b. $\{M \prec (H_\theta, \in, \mathbb{P}, p) : |M| = \omega \wedge (\exists q \leq p)(q \text{ is } M^\mathbb{P}\text{-generic})\}$ is club in $[H_\theta]^\omega$.
- c. For every countable $M \prec (H_\theta, \in, \mathbb{P}, p)$, there is $q \leq p$ that is $M^\mathbb{P}$ -generic.

Proof.

Implication (a $\Rightarrow$ c)

- 1° Let $M \prec (H_\theta, \in, \mathbb{P}, p)$ be arbitrary countable.
- 2° Inside H_θ , set $C := \{M \prec (H((2^{|\mathbb{P}|})^+), \in, \mathbb{P}, p) : |M| = \omega \wedge (\exists q \leq p)(q \text{ is } M^\mathbb{P}\text{-generic})\}$ is club.
- 3° Since $C \in M$, it is also club in M ; let $F \in M$ be an algebra on $H((2^{|\mathbb{P}|})^+)$ witnessing this fact.
- 4° Since $F \in M$, model M is closed for F ; consequently, $M \cap H((2^{|\mathbb{P}|})^+) \in C$.
- 5° Thus there is $q \leq p$ which is $(M \cap H((2^{|\mathbb{P}|})^+))^\mathbb{P}$ -generic.
- 6° Then q is clearly $M^\mathbb{P}$ -generic, as required.

Implication (c $\Rightarrow$ b)

7° Obvious.

Implication (b $\Rightarrow$ a)

- 8° Suppose that $S := \{M \prec (H((2^{|\mathbb{P}|})^+), \in, \mathbb{P}, p) : |M| = \omega \wedge (\forall q \leq p)(q \text{ is not } M^\mathbb{P}\text{-generic})\}$ is stationary.
- 9° Since $C := \{M \prec (H_\theta, \in, \mathbb{P}, p) : |M| = \omega \wedge (\exists q \leq p)(q \text{ is } M^\mathbb{P}\text{-generic})\}$ is club, we have $(S \upharpoonright H_\theta) \cap C \neq \emptyset$.
- 10° This is obviously a contradiction.

□

IX. **Definition.** A condition $p \in \mathbb{P}$ is *proper* if

$$\{M \prec (H((2^{|\mathbb{P}|})^+), \in, \mathbb{P}, p) : |M| = \omega \wedge (\exists q \leq p)(q \text{ is } M^\mathbb{P}\text{-generic})\}$$

is club. Poset $\mathbb{P}$ is *proper* if for every regular every $p \in \mathbb{P}$ is proper.

- X. **Proposition.** Let $\mathbb{P}$ be a poset and let $\theta > 2^{2^{|\mathbb{P}|}}$ be regular. The $\mathbb{P}$ is proper if and only if for every countable $M \prec (H_\theta, \in, \mathbb{P})$ and every $p \in M \cap \mathbb{P}$, there exists an $M^\mathbb{P}$ -generic condition $q \leq p$.

Proof. Clear. □

- XI. **Proposition.** Let $\mathbb{P}$ be proper and let $G \rightsquigarrow V^\mathbb{P}$. For every $X \subseteq V$ with $|X|^{V[G]} = \omega$, there is $Y \in V$ with $X \subseteq Y$ and $|Y|^V = \omega$.

Proof.

- 1° We work in V .
- 2° Suppose that $\Vdash \dot{f} : \omega \rightarrow V$ and let p be arbitrary. We want to find $q \leq p$ and $M \in [V]^\omega$ such that $q \Vdash \text{im } \dot{f} \subseteq M$.
- 3° Let θ be large enough regular.
- 4° Since $\mathbb{P}$ is proper, there is a countable $M \prec (H_\theta, \in, \mathbb{P}, p, \dot{f})$ such that there exists $q \leq p$ which is $M^\mathbb{P}$ -generic.
- 5° **Claim.** $q \Vdash \text{im } \dot{f} \subseteq M$

Proof.

- 1' Let $n < \omega$ be arbitrary.
- 2' There is $\dot{x} \in M^\mathbb{P}$ such that $q \Vdash \dot{x} = \dot{f}(n)$.
- 3' Since q is $M^\mathbb{P}$ -generic, we have $q \Vdash \dot{x} \in M$.
- 4' Thus, $q \Vdash \dot{f}(n) \in M$.

□

□

- XII. **Corollary.** A proper forcing does not collapse ω_1 .

- XIII. **Proposition.** Let $\mathbb{P}$ be a poset. Then the following are equivalent.

- a. $\mathbb{P}$ is proper.
- b. For every X and every stationary $S \subseteq [X]^\omega$, we have $\mathbb{P} \Vdash \ulcorner S \text{ is stationary} \urcorner$.

Proof.

Implication (a $\Rightarrow$ b)

- 1° Let $\Vdash \dot{F} : X^{<\omega} \rightarrow X$. We want to show $\Vdash (\exists M \in S)(\dot{F}[M^{<\omega}] \subseteq M)$.
- 2° Let $p \in \mathbb{P}$ be arbitrary. We want to find $q \leq p$ and $M \in S$ such that $q \Vdash \dot{F}[M^{<\omega}] \subseteq M$.
- 3° Let θ be regular large enough. There are club many countable $N \prec (H_\theta, \in, \mathbb{P}, p)$ for which there exists an $N^\mathbb{P}$ -generic $q \leq p$.

- 4° Since $S \upharpoonright H_\theta$ is stationary, there is a countable $N \prec (H_\theta, \in, \mathbb{P}, p, \dot{F})$ with $N \cap X \in S$ and an $N^\mathbb{P}$ -generic condition $q \leq p$.
- 5° We will show that $M := N \cap X$ and q are as required.
- 6° Let $x \in M^{<\omega}$ and let us show that $q \Vdash \dot{F}(x) \in M$.
- 7° Let $\dot{y} \in M^\mathbb{P}$ be such that $q \Vdash \dot{y} = \dot{F}(x)$.
- 8° $q \Vdash \dot{y} \in M$ since q is $M^\mathbb{P}$ -generic.
- 9° Thus, $q \Vdash \dot{F}(x) \in M$.

Implication (b $\Rightarrow$ a)

- 10° Let $\theta > 2^{|\mathbb{P}|}$ be regular and $p \in \mathbb{P}$. Let us assume towards a contradiction that the set S of all countable $M \prec (H_\theta, \in, \mathbb{P}, p)$ for which there is no $M^\mathbb{P}$ -generic $q \leq p$, is stationary.
- 11° Let $G \rightsquigarrow V^\mathbb{P}$ be arbitrary containing p and let us work in $V[G]$.
- 12° S remains stationary. In particular, there is a countable $N \prec (H_\theta^V[G], \in, \mathbb{P}, G)$ such that $M := N \cap H_\theta^V \in S$.
- 13° By elementarity, $N = M[G]$, while obviously $N \cap \text{Ord} = M \cap \text{Ord}$.
- 14° By the forcing theorem, there is $q \leq p$ in G such that $q \Vdash M[G] \cap \text{Ord} = M \cap \text{Ord}$.
- 15° In other words, $q \leq p$ is $M^\mathbb{P}$ -generic, which is a contradiction.

□

1.2 Examples

- I. **Proposition.** Let $\mathbb{P}$ be a poset and let $\theta > 2^{|\mathbb{P}|}$ be regular. Then $\mathbb{P}$ is c.c.c. if and only if for every countable $M \prec (H_\theta, \in, \mathbb{P})$, every condition in $\mathbb{P}$ is $M^\mathbb{P}$ -generic. In particular, every c.c.c. poset is proper.

Proof.

Implication ($\Rightarrow$)

- 1° Let $\mathbb{P}$ be c.c.c. and let $M \prec (H_\theta, \in, \mathbb{P})$ be arbitrary countable. We want to show that for every anti-chain $A \in M$, we have

$$\Vdash \mathbf{G} \cap A \cap M \neq \emptyset.$$

- 2° Since $\mathbb{P}$ is c.c.c, we have that A is countable.
- 3° Since $A \in M$ is countable, we have $A \subseteq M$.
- 4° Since $\Vdash \mathbf{G} \cap A \neq \emptyset$, we conclude $\Vdash \mathbf{G} \cap A \cap M \neq \emptyset$.

Implication ($\Leftarrow$)

- 5° Let A be an arbitrary anti-chain in $\mathbb{P}$. We want to show $|A| = \omega$.
- 6° Let $M \prec (H_\theta, \in, \mathbb{P}, A)$ be arbitrary countable. It suffices to show $A \subseteq M$.

- 7° Assume otherwise and let $p \in A - M$.
- 8° Let $G \rightsquigarrow V^{\mathbb{P}}$ be such that $G \ni p$. Then $G \cap A = \{p\}$.
- 9° This means $G \cap A \cap M = \emptyset$, so $p \in G$ cannot be $M^{\mathbb{P}}$ -generic. This is a contradiction with the assumption.

□

II. **Proposition.** Every σ -closed poset is proper.

Proof.

- 1° Let $p \in \mathbb{P}$ be arbitrary. Let θ be large enough regular and let $M \prec (H_\theta, \in, \mathbb{P}, p)$ be arbitrary. We want to find $M^{\mathbb{P}}$ -generic $q \leq p$.
- 2° Let $(D_n : n < \omega)$ enumerate all dense sets in M . We define a decreasing sequence $(p_n : n < \omega)$ of conditions in M by recursion.
- 3° $p_0 := p \in M$
- 4° $p_{n+1} \in D_n \cap M$ is chosen such that $p_{n+1} \leq p_n$. This is possible since $M \prec H_\theta$.
- 5° Let q satisfy $q \leq p_n$ for every $n < \omega$.
- 6° $q \Vdash \mathbf{G} \cap D_n \cap M \neq \emptyset$ for every $n < \omega$, implying that q is $M^{\mathbb{P}}$ -generic.

□

III. **Definition.** Let $\mathbb{P}$ be a proper poset and let $\mathbf{p} := (2^{|\mathbb{P}|})^+$. Poset $\mathbb{P}^*$ defined in the following manner.

- The conditions are of the form $q = (p_q, \mathcal{N}_q)$, where:
 - a. $p_q \in \mathbb{P}$
 - b. $\mathcal{N}_q$ is finite and linearly ordered by $\in$;
 - c. for every $N \in \mathcal{N}_q$, we have that $N \prec (H_{\mathbf{p}}, \in, \mathbb{P})$ and it is countable;
 - d. p_q is $N^{\mathbb{P}}$ -generic for every $N \in \mathcal{N}_q$.
- Order $r \leq q$ holds if $p_r \leq p_q$ and $\mathcal{N}_r \supseteq \mathcal{N}_q$.

IV. **Lemma.** Let $\mathbb{P}$ be a poset, let θ be large enough regular, let $M \prec (H_\theta, \in, \mathbb{P})$, and let $p \in \mathbb{P}$. Suppose that for every $D \in M$ which is dense in $\mathbb{P}$ and every $q \in D$ with $q \leq p$, there exists $r \in D \cap M$ which is compatible with q . Then p is $M^{\mathbb{P}}$ -generic.

Proof.

- 1° Let $D \in M$ be arbitrary dense. We need to show that $p \Vdash \mathbf{G} \cap D \cap M \neq \emptyset$.
- 2° For every $q_0 \leq p$, there is $q \in D$ with $q \leq q_0$.
- 3° Then there is $r \in D \cap M$ with $r \parallel q$. Consequently, $r \parallel q_0$.

- 4° Since every condition below p is compatible with an element in $D \cap M$, we have $p \Vdash \mathbf{G} \cap D \cap M \neq \emptyset$.

□

V. **Lemma.** If $\mathbb{P}$ is proper, then so is $\mathbb{P}^*$.

Proof.

- 1° Let θ be large enough, let $M \prec (H_\theta, \in, \mathbb{P})$, and let $q \in \mathbb{P}^* \cap M$.
 2° There is $p_{q_0} \leq p_q$ which is $M^{\mathbb{P}}$ -generic. Then clearly

$$q_0 := (p_{q_0}, \mathcal{N}_q \cup \{M \cap H_{\mathfrak{p}}\}) \in \mathbb{P}^*$$

and $q_0 \leq q$. Let us verify that q_0 is $M^{\mathbb{P}^*}$ -generic.

- 3° We will apply lemma 1-IV: let $D \in M$ be dense in $\mathbb{P}^*$, let $r \in D$ be such that $r \leq q_0$, and let us find $s \in D \cap M$ such that $s \parallel r$.
 4° Let $D' := \{p \in \mathbb{P} : (\exists s \in D)(\exists P \in \mathcal{N}_s)(p_s = p \wedge \mathcal{N}_s \cap P = \mathcal{N}_r \cap M)\} \in M$.
 Let $D'' := \{p \in \mathbb{P} : (\forall p' \in D')(p \perp p')\} \in M$.
 5° Since $D' \cup D'' \in M$ is pre-dense in $\mathbb{P}$ and since p_r is $M^{\mathbb{P}}$ -generic, we have $p_r \Vdash \mathbf{G} \cap ((D' \cup D'') \cap M) \neq \emptyset$. There are $p \leq p_r$ and $p' \in (D' \cup D'') \cap M$ such that $p \Vdash p' \in \mathbf{G}$.
 6° We have $p' \parallel p_r$. Since $p_r \in D'$, we have $p' \notin D''$.
 7° Thus, $p' \in D'$. This means that $(\exists s \in D)(\exists P \in \mathcal{N}_s)(p_s = p' \wedge \mathcal{N}_s \cap P = \mathcal{N}_r \cap M)$. By elementarity, since $p', \mathcal{N}_r \cap M \in M$, such an s can be chosen in M .
 8° We now have $s \parallel r$. To witness this, take any $p'' \leq p', p_r$ and consider condition $(p'', \mathcal{N}_s \cup \mathcal{N}_r)$.

□

2 Proper Forcing Axiom

I. **Definition.** PFA is the following statement: for every proper poset $\mathbb{P}$ and every family $(D_\alpha : \alpha < \omega_1)$ of dense sets, there is a filter $F \subseteq \mathbb{P}$ such that $(\forall \alpha < \omega_1)(F \cap D_\alpha \neq \emptyset)$.

II. **Proposition.** $\text{PFA} \implies \text{MA}_{\omega_1}$

Proof. Immediately from 1-I. □

III. **Theorem.** Suppose that κ is supercompact. Then there is a proper poset $\mathbb{P}$ such that $V^{\mathbb{P}} \models \text{PFA} + \kappa = \omega_2$.

3 Open Coloring Axiom

- I. **Notation.** A graph on set X is a symmetric irreflexive binary relation on X . The complete graph on X is $[X]^2 := \{(x, y) \in X^2 : x \neq y\}$. A graph on metric space X is open if it is an open subset of X^2 .
- II. **Definition.** *Open Coloring Axiom* OCA is the following statement: for every separable metric space X and every graph G on X , either

$$(\exists(Y_n : n < \omega))(X = \bigcup_{n < \omega} Y_n \wedge (\forall n < \omega)([Y_n]^2 \cap G = \emptyset))$$

or

$$(\exists Z \subseteq X)(|Z| > \omega \wedge [Z]^2 \subseteq G).$$

- III. **Theorem.** $\text{PFA} \implies \text{OCA}$

Proof. The rest of the section is devoted to the proof. $\square$

- IV. **Remark.** If X is a separable metric space, then $|X| \leq \mathfrak{c}$. We shall always assume that X is defined on an ordinal $\alpha \leq \mathfrak{c}$.

- V. **Definition.** Suppose that X is a separable metric space and that G is an open graph on X . Then poset $\mathbb{Q}_{X,G}$ is defined as follows.

- Conditions are pairs $q = (H_q, \mathcal{N}_q)$, such that:
 - a. $H_q \in [X]^{<\omega}$ and $[H_q]^2 \subseteq G$;
 - b. $\mathcal{N}_q$ if finite and linearly ordered by $\in$;
 - c. every $N \in \mathcal{N}_q$ is countable and $N \prec (H(\mathfrak{c}^+), \in, X, G)$;
 - d. $(\forall x, y \in H_q)(\exists N \in \mathcal{N}_q)(x \neq y \implies |N \cap \{x, y\}| = 1)$;
 - e. $(\forall N \in \mathcal{N}_q)(\forall x \in H_q - N)(\forall E \subseteq X)([E]^2 \cap G = \emptyset \wedge E \in N \implies x \notin E)$.
- Order is defined such that $p \leq q$ if and only if $H_p \supseteq H_q$ and $\mathcal{N}_p \supseteq \mathcal{N}_q$.

- VI. **Remark.** For $x \in H_q$, let $\#x := \{N \in \mathcal{N}_q : x \notin N\}$. By [V.d](#), we have that $\# : H_q \rightarrow |\mathcal{N}_q| + 1$ is an injection. Let $x_q(i)$ denote the i^{th} element in the order on H_q pulled back by $\#$.

- VII. **Lemma.** Let X be a separable metric space, let $S \subseteq X$ be dense countable, let G be an open graph on X , and let $P \prec (H(\mathfrak{c}^+), \in, X, G, S)$ be countable. Suppose that:

- a. $A \subseteq X^n$ and $A \in P$;
- b. $x \in \overline{A}$ and $x \upharpoonright (n-1) \in P$;
- c. $(\forall E \subseteq X)([E]^2 \cap G = \emptyset \wedge E \in P \implies x(n-1) \notin E)$.

Then there is an open set O such that:

- i. O is a ball with the center in S and a rational radius;

- ii. $x(n-1) \notin O$
- iii. $(\forall y \in O)((x(n-1), y) \in G)$
- iv. $x \upharpoonright (n-1) \in \overline{\{y \upharpoonright (n-1) : y \in A, y(n-1) \in O\}}$

Proof.

- 1° Let $(W_i : i < \omega) \in P$ be a local base for the topological space X^{n-1} at $x \upharpoonright (n-1)$.
- 2° Let $E_0 := \bigcap_{i < \omega} \overline{\{y(n-1) : y \in A, y \upharpoonright (n-1) \in W_i\}} \in P$.
- 3° Let $E := \{z \in E_0 : (\forall y \in E_0)((y, z) \notin G)\} \in P$.
- 4° Clearly, $x(n-1) \in E_0$.
- 5° By [c](#), $x(n-1) \notin E$.
- 6° Consequently, there is $y \in E_0$ such that $(y, x(n-1)) \in G$.
- 7° Since $y \neq x(n-1)$ and G is open, there is a ball O with the center in S and a rational radius such that $y \in O$, $x(n-1) \notin O$ and $\{x(n-1)\} \times O \subseteq G$.
- 8° **Claim.** $x \upharpoonright (n-1) \in \overline{\{y \upharpoonright (n-1) : y \in A, y(n-1) \in O\}}$

Proof.

- 1' Let $i < \omega$ be arbitrary and let us show that

$$W_i \cap \{y \upharpoonright (n-1) : y \in A, y(n-1) \in O\} \neq \emptyset. \quad (1)$$

- 2' Since $y \in E_0$, we have $y \in \overline{\{z(n-1) : z \in A, z \upharpoonright (n-1) \in W_i\}}$.
- 3' Since $O \ni y$ is open, there is $z \in A$ with $z \upharpoonright (n-1) \in W_i$ such that $z(n-1) \in O$.
- 4' z witnesses [\(1\)](#).

□

□

VIII. Lemma. Let θ be large enough regular and let $M \prec (H_\theta, \in, \mathbb{Q}_{X,G})$ be countable. For every $q \in \mathbb{Q}_{X,G} \cap M$, we have $q_0 := (H_q, \mathcal{N}_q \cup \{M \cap H(\mathfrak{c}^+)\}) \in \mathbb{Q}_{X,G}$, $q_0 \leq q$, and q_0 is $M^{\mathbb{Q}_{X,G}}$ -generic.

Proof.

- 1° $q \in \mathbb{Q}_{X,G}$ and $q_0 \leq q$ is readily verifiable. (For [V.e](#), note that since $H_q \in M$, then every subset of this finite set belongs to and is contained in M .)
- 2° We will show that q_0 is $M^{\mathbb{Q}_{X,G}}$ -generic by applying lemma [1-IV](#).
- 3° Let $D \in M$ be dense containing q_0 and let $r \in D$ be arbitrary. We want to find $s \in D \cap M$ compatible with r .
- 4° Let $N := M \cap H(\mathfrak{c}^+) \prec H(\mathfrak{c}^+)$, let $r_0 := (H_r \cap N, \mathcal{N}_r \cap N)$, and let $i_0 := |H_{r_0}|$. We have $r_0 \in N$.

- 5° Let $n := |H_r|$ and let $S \in N$ be a countable dense subset of X .
- 6° **Claim.** There are open disjoint sets $(U_i : i < n) \in N$ such that $x_r(i) \in U_i$ and $U_i \times U_j \subseteq G$ ($i, j < n, i \neq j$).

Proof.

- 1' Since G is open, there are disjoint open sets $(\tilde{U}_i : i < n)$ such that $x_r(i) \in \tilde{U}_i$ and $\tilde{U}_i \times \tilde{U}_j \subseteq G$ ($i, j < n, i \neq j$).
- 2' There are $(x_i : i < n) \in S^n$ and rationals $(y_i : i < n)$ such that $x_r(i) \in B(x_i, y_i) \subseteq \tilde{U}_i$ ($i < n$).
- 3' Since $S \subseteq N$, we have $B(x_i, y_i) \in N$.
- 4' Thus, $U_i := B(x_i, y_i)$ are as required. □

- 7° Let D' consist of all $s \leq r_0$ satisfying:

- a. $(\exists P \in \mathcal{N}_s)(H_s \cap P = H_{r_0} \wedge \mathcal{N}_s \cap P = \mathcal{N}_{r_0})$
- b. $|H_s| = n$
- c. $x_s(i) \in U_i$ for all $i < n$

Note that $D' \in N$.

- 8° **Claim.** There are open sets $(O_i : i_0 \leq i < n) \in N$ such that:

- a. $(\forall i \in [i_0, n])(\forall y \in O_i)((x_r(i), y) \in G)$
- b. $(\exists s \in D')(\forall i \in [i_0, n])(x_s(i) \in O_i)$.

Proof.

- 1' We proceed by reverse recursion on $i \in [i_0, n]$ to define sets $A_i \subseteq X^n$ and open sets O_i (when $i < n$).
- 2' Let $A_n := \{x_s : s \in D'\}$. We have $A_n \in N$ and $x_r \in \overline{A_n}$.
- 3' Assume recursively that A_i ($i \in (0, n]$) has been defined and that it satisfies $A_i \subseteq X^n$, $A_i \in N$, and

$$x_r \restriction i \in \overline{\{y \restriction i : y \in A_i\}}.$$

- 4' Let $P \in \mathcal{N}_r$ be the last model with $x_r(i-1) \notin P$. Note that $P = N$ or $P \ni N$.
- 5' Since $A_i \in N$, we have $A_i \in P$.
- 6' $x_r \restriction (i-1) \in P$
- 7' For every $E \in P$ such that $E \subseteq X$ and $[E]^2 \cap G = \emptyset$, we have $x_r(i-1) \notin E$. (This follows from [V.e](#) and the fact that $x_r(i-1) \notin P$.)
- 8' By lemma [3-VII](#), there is O_{i-1} such that:
- a. O_{i-1} is a ball with the center in S and a rational radius;
 - b. $x_r(i-1) \notin O_{i-1}$
 - c. $(\forall y \in O_{i-1})((x_r(i-1), y) \in G)$
 - d. $x_r \restriction (i-1) \in \overline{\{y \restriction (i-1) : y \in A_i, y(i-1) \in O_{i-1}\}}$.

- 9' Since $O_{i-1} \in N$, we have $A_{i-1} := \{y \in A_i : y(i-1) \in O_{i-1}\} \in N$.
- 10' This concludes the recursion. Property 8°a is obvious.
- 11' For 8°b, it suffices to verify $A_{i_0} \neq \emptyset$.
- 12' Note that by 8'd, we have $x_r \restriction i_0 \in \overline{\{y \restriction i_0 : y \in A_{i_0+1}, y(i_0) \in O_{i_0}\}} = \{y \restriction i_0 : y \in A_{i_0}\}$. This means $A_{i_0} \neq \emptyset$.

□

9° By 8°b and elementarity, there is $s \in D' \cap M \subseteq D \cap M$ such that $(\forall i \in [i_0, n))(x_s(i) \in O_i)$.

10° **Claim.** $s \parallel r$

Proof.

- 1' It suffices to show that $(H_r \cup H_s, \mathcal{N}_r \cup \mathcal{N}_s)$ is a condition.
- 2' Let us verify that $[H_r \cup H_s]^2 \subseteq G$.
- 3' For $i \neq j$, we have $(x_r(i), x_s(j)) \in U_i \times U_j \subseteq G$.
- 4' For $i = j$ and $x_r(i) \neq x_s(j)$, we have $i \in [i_0, n)$. This follows from the fact that $x_r \restriction i_0 = x_s \restriction i_0$.
- 5' Since $x_s(i) \in O_i$, property 8°a implies $(x_r(i), x_s(i)) \in G$.
- 6' To see that $\mathcal{N}_r \cup \mathcal{N}_s$ is linearly ordered by $\in$, note that $\mathcal{N}_r$ consists of the interval of models $\mathcal{N}_{r_0}$ followed by the interval of models that are equal or contain N . On the other hand, since $s \in M$, set $\mathcal{N}_s$ consists of the interval $\mathcal{N}_{r_0}$ followed by another interval of models, all which are contained in N .
- 7' It remains to verify that for every $x, y \in H_r \cup H_s$ with $x \neq y$, there exists $P \in \mathcal{N}_r \cup \mathcal{N}_s$ such that $|P \cap \{x, y\}| = 1$.
- 8' We only need to consider the case $x \in H_r - H_{r_0}$ and $y \in H_s - H_{r_0}$.
- 9' We have $x \notin N$, while $y \in H_s \subseteq N$, which suffices considering that $N \in \mathcal{N}_r$.

□

□

IX. **Corollary.** $\mathbb{Q}_{X,G}$ is proper.

X. *Proof of 3-III.*

1° Assume

$$\neg(\exists(Y_n : n < \omega))(X = \bigcup_{n < \omega} Y_n \wedge (\forall n < \omega)([Y_n]^2 \cap G = \emptyset)). \quad (2)$$

- 2° Let θ be large enough regular and let $M \prec (H_\theta, \in, \mathbb{Q}_{X,G})$ be countable. By (2), there is $x \in X$ such that $x \notin E$ for every $E \subseteq X$ with $[E]^2 \cap G = \emptyset$ and $x \in M$.
- 3° Let $q := (\{x\}, \{M \cap H(\mathfrak{c}^+)\})$. By 3-VIII, condition q is $M^{\mathbb{Q}_{X,G}}$ -generic.

4° **Claim.** $q \Vdash |\mathbf{G}| > \omega$.

Proof.

- 1' Assume otherwise and let $H \rightsquigarrow V^{\mathbb{Q}_{X,G}}$ be such that $H \ni q$ and $|H| \leq \omega$.
- 2' Since q is $M^{\mathbb{Q}_{X,G}}$ -generic, we have $M[H] \cap V = M$ and $M[H] \prec H_\theta^{V[H]}$.
- 3' Since $H \in M[H]$ and it is countable, we conclude $H \subseteq M[H] \cap V = M$.
- 4' This is a contradiction, since $x \in H - M$.

□

5° There is a name $\dot{f}$ such that $q \Vdash \dot{f} : \omega_1 \xrightarrow{1-1} \mathbf{G}$.

6° For $\alpha < \omega_1$, let A_α be a maximal anti-chain below q , such that for every $r \in A_\alpha$, there is $x_{\alpha,r} \in X$ such that $r \Vdash \dot{f}(\alpha) = x_{\alpha,r}$.

7° By PFA, there is a filter $F \subseteq \mathbb{Q}_{X,G}$ such that $(\forall \alpha < \omega_1)(F \cap A_\alpha \neq \emptyset)$.

8° For $\alpha < \omega_1$, we denote $r_\alpha \in F \cap A_\alpha$ and $y_\alpha := x_{\alpha,r_\alpha}$.

9° Let $Z := \{y_\alpha : \alpha < \omega_1\} \subseteq X$.

10° For $\alpha \neq \beta$, there is $s_{\alpha,\beta} \leq r_\alpha, r_\beta$ such that $s \in F$. Then $s_{\alpha,\beta} \Vdash y_\alpha = \dot{f}(\alpha) \neq \dot{f}(\beta) = y_\beta$ and consequently, $y_\alpha \neq y_\beta$.

11° In particular, we have $|Z| = \omega_1$.

12° Since $\Vdash [\mathbf{G}]^2 \subseteq G$, for $\alpha \neq \beta$, we have $s_{\alpha,\beta} \Vdash (y_\alpha, y_\beta) = (\dot{f}(\alpha), \dot{f}(\beta)) \in [\mathbf{G}]^2 \subseteq G$ and consequently, $(y_\alpha, y_\beta) \in G$.

13° Thus, $[Z]^2 \subseteq G$.

□

4 P-ideal Dichotomy

- I. **Notation.** $A \subseteq_* B$ means $|A - B| < \omega$.
- II. **Definition.** An ideal I on a set S is a *p-ideal* if $[S]^{<\omega} \subseteq I \subseteq [S]^\omega$ and every countable subset of I has an upper $\subseteq_*$ -bound inside I .
- III. **Definition.** *P-ideal Dichotomy* PID is the following statement: for every p-ideal I on a set S , either

$$(\exists (S_n : n < \omega))(S = \bigcup_{n < \omega} S_n \wedge (\forall n < \omega)([S_n]^\omega \cap I \subseteq [S_n]^{<\omega}))$$

or

$$(\exists Z \subseteq S)(|Z| > \omega \wedge [Z]^\omega \subseteq I).$$

- IV. **Theorem.** $\text{PFA} \implies \text{PID}$

Proof. The remainder of the section is devoted to the proof of this theorem. $\square$

- V. **Definition.** If I is a p-ideal on S , then we define its *orthogonal ideal* as

$$I^\perp := \{X \subseteq S : (\forall Y \in I)(|X \cap Y| < \omega)\}.$$

- VI. **Remark.** In what follows, we will always assume that S is an infinite cardinal. Once it is fixed, we will denote $\mathfrak{s} := (S^\omega)^+$. Once a p-ideal I has been fixed, we also fix an assignment $M \mapsto I_M$, where $M \prec H_{\mathfrak{s}}$ is countable and $I_M \in I$ is an upper $\subseteq_*$ -bound for $I \cap M$.

- VII. **Definition.** For a p-ideal I on S , we define poset $\mathbb{Q}_I$ as follows.

- Conditions are pairs $q = (Z_q, \mathcal{N}_q)$ satisfying:
 - a. $Z_q \in [S]^{<\omega}$
 - b. $\mathcal{N}_q$ is finite and linearly ordered by $\in$;
 - c. $(\forall N \in \mathcal{N}_q)(|N| = \omega \wedge N \prec (H_{\mathfrak{s}}, \in, I))$
 - d. $(\forall x, y \in Z_q)(\exists N \in \mathcal{N}_q)(x \neq y \implies |N \cap \{x, y\}| = 1)$
 - e. $(\forall N \in \mathcal{N}_q)(\forall x \in Z_q - N)(\forall X \in I^\perp \cap N)(x \notin X)$.
- Order $p \leq q$ holds if:
 - f. $Z_p \supseteq Z_q$
 - g. $\mathcal{N}_p \supseteq \mathcal{N}_q$
 - h. $(\forall N \in \mathcal{N}_q)((Z_p - Z_q) \cap N \subseteq I_N)$.

- VIII. **Remark.** For $x \in Z_q$, let $\#x := \{N \in \mathcal{N}_q : x \notin N\}$. By VII.d, we have that $\# : Z_q \rightarrow |\mathcal{N}_q| + 1$ is an injection. Let $t_q(i)$ denote the i^{th} element in the order on Z_q pulled back by $\#$.

IX. **Proposition.** Let θ be large enough regular, let $M \prec (H_\theta, \in, \mathbb{Q}_I)$ be countable, let $q \in \mathbb{Q}_I \cap M$ be arbitrary. Then $q_0 := (Z_q, \mathcal{N}_q \cup \{M \cap H_s\}) \in \mathbb{Q}_I$, $q_0 \leq q$, and q_0 is $M^{\mathbb{Q}_I}$ -generic.

Proof.

- 1° $q_0 \in \mathbb{Q}_I$ and $q_0 \leq q$ is obvious. Let us verify $M^{\mathbb{P}}$ -genericity.
- 2° Let $r \leq q_0$ and let $D \in M$ be dense such that $r \in D$. We want to find $s \in D \cap M$ such that $r \parallel s$.
- 3° Let $n := |Z_r|$.
- 4° Let $N := M \cap H_s$, let $r_0 := (Z_r \cap N, \mathcal{N}_r \cap N)$, and let $i_0 := |Z_{r_0}|$.
- 5° Let D' be the set of all $s \leq r_0$ such that $(\exists P \in \mathcal{N}_s)(r_0 = (Z_s \cap P, \mathcal{N}_s \cap P))$ and $|Z_s| = n$. Note that $D' \in M$.
- 6° *Notation.* We denote $\pi_i : S^n \rightarrow S^i : t \mapsto t \upharpoonright i$ for $i \leq n$.
- 7° **Claim.** There is a set $U \subseteq S^n$ with the following properties:
 - a. $U \subseteq \{t_s : s \in D'\}$ and $U \in N$;
 - b. $t_r \in U$
 - c. for every $i \in [i_0, n)$ and every $z \in \pi_i[U]$, we have

$$\{x \in S : z \frown x \in \pi_{i+1}[U]\} \in (I^\perp)^+.$$

Proof.

- 1' By reverse recursion on $i \in [i_0, n]$, we define sets $T_i \subseteq S^n$.
- 2' We set $T_n := \{t_s : s \in D'\} \in N$.
- 3' Assume recursively that $T_i \subseteq S^n$ with $T_i \in N$ ($i \in (i_0, n]$) has been defined.
- 4' We define set $T'_{i-1} \subseteq S^{i-1}$ by setting
$$z \in T'_{i-1} : \iff z \in \pi_{i-1}[T_i] \wedge \{x \in S : z \frown x \in \pi_i[T_i]\} \in (I^\perp)^+.$$
- 5' Then we define $T_{i-1} := \{y \in T_i : y \upharpoonright (i-1) \in T'_{i-1}\} \in N$.
- 6' This concludes the recursion. Let $U := T_{i_0}$.
- 7' Clearly $U \subseteq \{t_s : s \in D'\}$ and $U \in N$.
- 8' **Claim.** $t_r \in U$

Proof.

- 1'' By reverse induction on $i \in [i_0, n]$, we show $t_r \in T_i$.
- 2'' $t_r \in T_n$ is clear.
- 3'' Suppose inductively that $t_r \in T_i$ and let us show $t_r \in T_{i-1}$. We need to verify $t_r \upharpoonright (i-1) \in T'_{i-1}$.
- 4'' In other words, we need to verify

$$\{x \in S : (t_r \upharpoonright (i-1)) \frown x \in \pi_i[T_i]\} \notin I^\perp.$$

5'' Let $P \in \mathcal{N}_q$ be largest such that $t_r(i-1) \notin P$. We have $P = N$ or $P \ni N$.

6'' Since $t_r \in T_i$, we conclude $t_r \upharpoonright i \in \pi_i[T_i]$.

7'' Consequently, $t_r(i-1) \in \{x \in S : (t_r \upharpoonright (i-1)) \frown x \in \pi_i[T_i]\}$

8'' By property VII.e applied to model P , we have $\{x \in S : (t_r \upharpoonright (i-1)) \frown x \in \pi_i[T_i]\} \notin I^\perp$.

□

9' **Claim.** For every $i \in [i_0, n)$, $\{y \in T_{i+1} : y \upharpoonright i \in \pi_i[T_{i_0}]\} \subseteq T_{i_0}$.

Proof.

1'' We reason by induction on i .

2'' By 7°5', we have $\pi_{i_0}[T_{i_0}] \subseteq T'_{i_0}$.

3'' Stage $i = i_0$ follows now from 7°9'2'' and 7°5'.

4'' Let us consider inductive step $i + 1$.

5'' Let $y \in T_{i+2}$ be arbitrary with $y \upharpoonright (i+1) \in \pi_{i+1}[T_{i_0}]$.

6'' It suffices to show $y \in T_{i+1}$, for then we may apply the IH directly to y .

7'' We need to verify $y \upharpoonright (i+1) \in T'_{i+1}$ (cf. 7°5').

8'' By 7°5', we have $\pi_{i+1}[T_{i_0}] \subseteq \pi_{i+1}[T_{i+1}] \subseteq T'_{i+1}$.

9'' Thus, $y \upharpoonright (i+1) \in \pi_{i+1}[T_{i_0}] \subseteq T'_{i+1}$.

□

10' **Claim.** For every $i \in [i_0, n)$ and every $z \in \pi_i[U]$, we have

$$\{x \in S : z \frown x \in \pi_{i+1}[U]\} \in (I^\perp)^+.$$

Proof.

1'' $z \in \pi_i[U] \subseteq \pi_i[T_i] \subseteq T'_i$.

2'' By 7°4', we have

$$\{x \in S : z \frown x \in \pi_{i+1}[T_{i+1}]\} \in (I^\perp)^+.$$

3'' By 7°9', if $z \frown x \in \pi_{i+1}[T_{i+1}]$ and $z \in \pi_i[U]$, then $z \frown x \in \pi_{i+1}[U]$.

4'' Consequently,

$$\{x \in S : z \frown x \in \pi_{i+1}[T_{i+1}]\} \subseteq \{x \in S : z \frown x \in \pi_{i+1}[U]\}.$$

5'' By 7°10'2'', we conclude

$$\{x \in S : z \frown x \in \pi_{i+1}[U]\} \in (I^\perp)^+.$$

□

11' Thus, U is as required.

□

8° **Claim.** There is $t \in U \cap M$ such that $t \restriction i_0 = t_r \restriction i_0$ and

$$\text{im } t \subseteq \bigcap_{P \in \mathcal{N}_r - N} I_P.$$

Proof.

- 1' We define $t \restriction i \in \pi_i[U] \cap M$ by recursion on $i \in [i_0, n]$.
- 2' $t \restriction i_0 := t_r \restriction i_0 \in \pi_{i_0}[U] \cap M$
- 3' Assume recursively that $t \restriction i \in \pi_i[U] \cap M$ has been defined.
- 4' By 7°c, we have $\{x \in S : (t \restriction i) \cap x \in \pi_{i+1}[U]\} \in (I^\perp)^+ \cap M$.
- 5' Consequently, there is an infinite $H \in I \cap M$ such that for every $x \in H$, we have $(t \restriction i) \cap x \in \pi_{i+1}[U]$.
- 6' For every $P \in \mathcal{N}_r - N$, we have $H \in P$. Consequently, $H \subseteq_* I_P$.
- 7' We have $H \cap \{P : P \in \mathcal{N}_r - N\} \neq \emptyset$. Let $t(i)$ be an element of this set.
- 8' Note that $t(i) \in H \subseteq M$. Thus, $t(i)$ is as required.

□

9° There is $s \in D' \cap M$ such that $t = t_s$.

10° **Claim.** $s \parallel r$

Proof.

- 1' It is easily seen that $(Z_r \cup Z_s, \mathcal{N}_r \cup \mathcal{N}_s)$ is a condition. Let us verify that it is $\leq r, s$.
- 2' We need to establish that:
 - a. $(\forall P \in \mathcal{N}_r)((Z_r \cup Z_s) - Z_r) \cap P \subseteq I_P$
 - b. $(\forall P \in \mathcal{N}_s)((Z_r \cup Z_s) - Z_s) \cap P \subseteq I_P$.
- 3' Equivalently, we need to establish:
 - a. $(\forall P \in \mathcal{N}_r)((Z_s - Z_{r_0}) \cap P \subseteq I_P)$
 - b. $(\forall P \in \mathcal{N}_s)((Z_r - Z_{r_0}) \cap P \subseteq I_P)$.
 (The reason for this equivalence is that $Z_s \subseteq N$, while $Z_r \cap N = Z_{r_0}$.)
- 4' Let us first verify 10°3'a.
- 5' If $P \in \mathcal{N}_{r_0}$, we have $(Z_s - Z_{r_0}) \cap P = \emptyset$.
- 6' If $P \in \mathcal{N}_r - \mathcal{N}_{r_0} = \mathcal{N}_r - N$, we have

$$(Z_s - Z_{r_0}) \cap P \subseteq \text{im } t \subseteq I_P.$$

- 7' To verify 10°3'b, we observe that $(Z_r - Z_{r_0}) \cap P = \emptyset$ for every $P \in \mathcal{N}_s$.

□

11° Thus, s is as required in 2°.

□

X. **Corollary.** Poset $\mathbb{Q}_I$ is proper.

XI. *Proof of 4-IV.*

1° Let $\dot{Z}$ be such that $\Vdash \dot{Z} = \bigcup \{Z_p : p \in \mathbb{G}\}$.

2° **Claim.** $\Vdash [\dot{Z}]^\omega \subseteq I$

Proof.

1' We work in $V[G]$ for a generic G . Let $Z := \dot{Z}^G$.

2' Let $X \in [Z]^\omega$ be arbitrary. We want to show that $X \in I$.

3' There is $Y \in V$ such that $X \subseteq Y$ and $|Y|^V = \omega$.

4' **Subclaim.** Set $D := \{p \in \mathbb{Q}_I : (\exists N \in \mathcal{N}_p)(Y \in N)\}$ is dense.

Proof.

1'' Let $q \in \mathbb{Q}_I$ be arbitrary and let us find $q \in D$ with $q \leq p$.

2'' Let θ be large enough regular and let $M \prec (H_\theta, \in, \mathbb{Q}_I, Y)$ be countable.

3'' By 4-IX, $p := (Z_q, \mathcal{N}_q \cup \{M \cap H_s\}) \in \mathbb{Q}_I$ and $p \leq q$.

4'' Since $Y \in M \cap H_s \in \mathcal{N}_p$, we conclude $p \in D$. □

5' Thus, there is $p \in G \cap D$.

6' We have $N \in \mathcal{N}_p$ such that $Y \in N$. Thus, $X \subseteq Y \subseteq N$.

7' We are done if we verify $X \subseteq Z_p \cup I_N \in I$. Let $x \in X - Z_p$ be arbitrary and let us show $x \in I_N$.

8' Since $X \subseteq Z$, there is $q \in G$ such that $q \leq p$ and $x \in Z_q$.

9' Since $q \leq p$, we have $x \in (Z_q - Z_p) \cap N \subseteq I_N$. □

3° Let us assume that

$$\neg(\exists(S_n : n < \omega))(S = \bigcup_{n < \omega} S_n \wedge (\forall n < \omega)([S_n]^\omega \cap I \subseteq [S_n]^{<\omega})).$$

4° Observe that for $X \subseteq S$, we have

$$X \in I^\perp \iff (\forall Y \in I)(|X \cap Y| < \omega) \tag{3}$$

$$\iff P(X) \cap I \subseteq [X]^{<\omega} \tag{4}$$

$$\iff [X]^\omega \cap I \subseteq [X]^{<\omega}. \tag{5}$$

5° Thus,

$$\neg(\exists(S_n : n < \omega))(S = \bigcup_{n < \omega} S_n \wedge (\forall n < \omega)(S_n \in I^\perp)).$$

6° Let θ be large enough regular and let $M \prec (H_\theta, \in, \mathbb{Q}_I)$ be countable. By 5°, there is $x \in S - \bigcup \{X : X \in I^\perp \cap M\}$.

7° By 4-IX, $q := (\{x\}, \{M \cap H_\mathfrak{s}\})$ is $M^{\mathbb{Q}_I}$ -generic.

8° **Claim.** $q \Vdash |\dot{Z}| > \omega$

Proof.

1' Assume otherwise and let $G \rightsquigarrow V^{\mathbb{Q}_I}$ be such that $G \ni q$ and inside $V[G]$, we have $|\dot{Z}^G| \leq \omega$. Let us denote $Z := \dot{Z}^G$.

2' Since q is $M^{\mathbb{Q}_I}$ -generic, we have $M[G] \prec H_\theta^{V[G]}$ and $M[G] \cap V = M$.

3' $Z = \{Z_p : p \in G\} \in M[G]$.

4' Since $|Z| \leq \omega$, we have $Z \subseteq M[G] \cap V = M$.

5' This is a contradiction since $x \in Z - M$.

□

9° Let $\dot{f}$ be such that $q \Vdash \dot{f} : \omega_1 \xrightarrow{1-1} \dot{Z}$.

10° By 2°, there is $\dot{g}$ such that

$$q \Vdash \dot{g} : \omega_1 \rightarrow I \wedge (\forall \alpha < \omega_1)(\dot{f}[\alpha] \subseteq \dot{g}(\alpha)).$$

11° For $\alpha < \omega_1$, let A_α be a maximal anti-chain below q such that for every $r \in A_\alpha$, there are $x_{\alpha,r} \in S$ and $X_{\alpha,r} \in I$ such that

$$r \Vdash \dot{f}(\alpha) = x_{\alpha,r} \wedge \dot{g}(\alpha) = X_{\alpha,r}.$$

12° By PFA, there is a filter $F \subseteq \mathbb{Q}_I$ such that $(\forall \alpha < \omega_1)(F \cap A_\alpha \neq \emptyset)$. For $\alpha < \omega_1$, let $r_\alpha \in F \cap A_\alpha$, let $y_\alpha := x_{\alpha,r_\alpha}$ and $Y_\alpha := X_{\alpha,r_\alpha}$, and let $W := \{y_\alpha : \alpha < \omega_1\}$.

13° **Claim.** $|W| = \omega_1$ and $[W]^\omega \subseteq I$.

Proof.

1' For $\alpha \neq \beta$, there is $s \leq r_\alpha, r_\beta$. Then $s \Vdash y_\alpha = \dot{f}(\alpha) \neq \dot{f}(\beta) = y_\beta$ and consequently, $y_\alpha \neq y_\beta$.

2' We conclude $|W| = \omega_1$.

3' Suppose $U \in [W]^\omega$. Then there is $\alpha < \omega_1$ such that $U \subseteq \{y_\beta : \beta < \alpha\} =: U'$.

4' For $\beta < \alpha$, let $s \leq r_\alpha, r_\beta$. We have $s \Vdash y_\beta = \dot{f}(\beta) \in \dot{g}(\alpha) = Y_\alpha$ and consequently, $y_\beta \in Y_\alpha$.

5' Thus, $U' \subseteq Y_\alpha \in I$ and consequently, $U \in I$.

□

□